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The “Sustainable Economy and Ecotechnology” (SEE) book series acts as a timely initiative within the modern academic sphere, fostering a strong connection between technology, research, policy making, and a successful practical implementation for a sustainable future. In an age marked by increasingly urgent environmental challenges and by the critical importance of sustainable development. The series editor is dedicated to steering positive efforts to provide a strong platform for the dissemination of high-quality research focusing on sustainable business, technology, and ecological management and conservation. Through the contributions of experts in the field, “SEE” endeavors to create a transition toward sustainable economic models and technologically driven ecological awareness. The “SEE” series welcomes submissions of selected papers from conference proceedings, monographs, textbooks, and edited volumes, aiming to cultivate a comprehensive understanding of sustainable development by shining a spotlight on a diverse array of topics that collectively contribute to the advancement of a Sustainable Economy and Eco-technology. The scope of this series includes, but is not limited to: Sustainable business practices and management strategies and digitalization, circular economy principles and benefits, climate change impact on economy and business operation, renewable energy and its impact in driving economic growth, role of technology in ecological and environmental preservation, sustainable agriculture, sustainable economics, sustainability in digital business management, finance and investment, sustainable supply chain management: impacting marketing, finance, production, manufacturing and distribution, Implications of economic development on the environment, sustainable mobility and its economic impacts, sustainable marketing strategies and its socio-economic implications, sustainable urban and rural development impacts on economy and more topics related to business investment, economy, environment, and technology.

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Preface

This book comprises the proceedings of the Research Conclave and Conference conducted by Thiagarajar College of Engineering, Madurai, India, in collaboration with Anurag University, India, on May 16–17, 2025, with the central theme of Sustainability. The conclave serves as a confluence of scholars, innovators, and thought leaders dedicated to developing sustainable solutions that address environmental, technological, and societal challenges.

This book focuses on applying intelligent computing techniques for sustainable development in precision agriculture, digital healthcare, renewable energy, smart cities, and autonomous systems. These areas are highly relevant in the current context, particularly for achieving sustainable development goals such as zero hunger, good health and well-being, and industry innovation and infrastructure.

It bridges the gap between artificial intelligence and sustainable solutions, providing readers with cutting-edge research and practical applications aimed at addressing environmental challenges. By highlighting real-world case studies, the book demonstrates how intelligent systems can drive transformation to meet global sustainability goals. Through its interdisciplinary approach, it encourages collaboration across sectors to create sustainable solutions.

It will serve as an invaluable resource for academicians, scientists, researchers, and industry experts worldwide, delving into research and development in the agriculture, healthcare, automation, and energy sectors.

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